BUILDTHESIMULATION / NEURAL INTERFACE DRAWING SERIES

NEUROPIXELS 1.0

MULTIVIEW 3D ENGINEERING STUDY / DOCUMENTED SHANK + ILLUSTRATION-DERIVED HEAD

NPX–001

REV 01 / SHEET 01 OF 02

25 OCTOBER 2026

A Isometric reconstruction

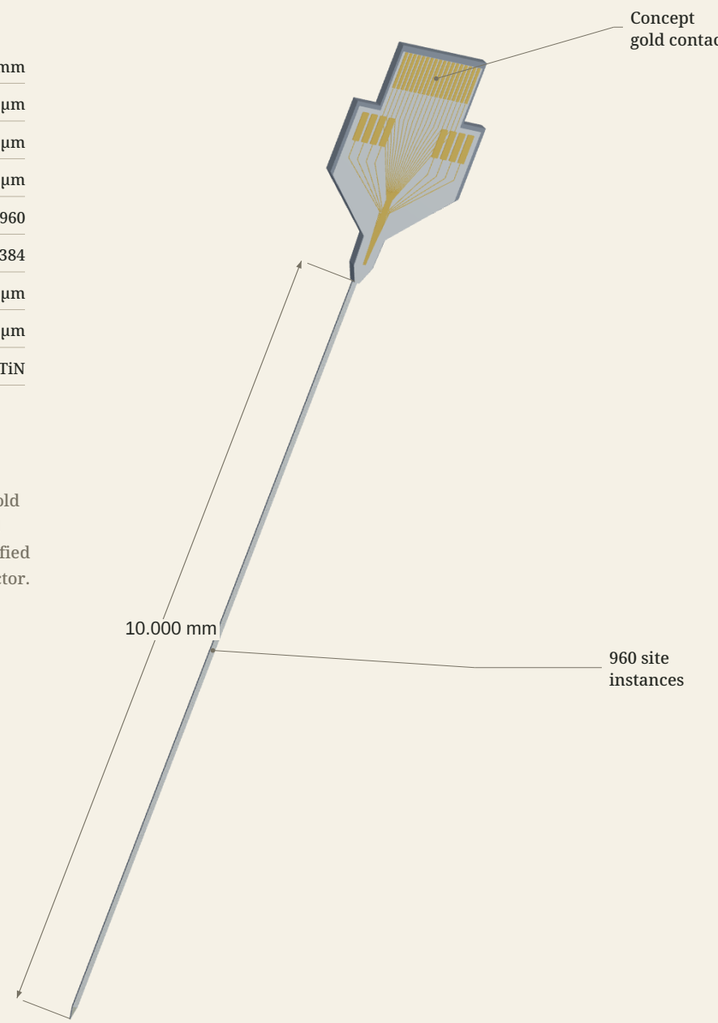
True aspect ratio; no shank thickening

DOCUMENTED NOMINALS

Shank length	10.000 mm
Shank width	70 µm
Shank thickness	24 µm
Tip length	175 µm
Recording sites	960
Recording channels	384
Site footprint	12 × 12 µm
Row / column pitch	20 / 16 µm
Contact material	Porous TiN

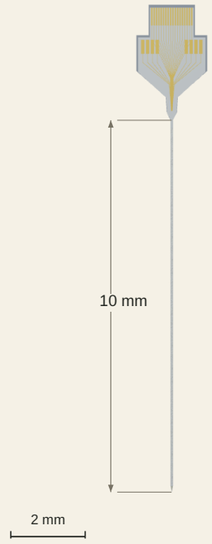
MODEL STATUS

Head outline, pad layout and gold routing reproduce the supplied illustration. They are not a verified Neuropixels production connector.



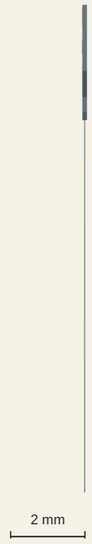
B Front elevation

Orthographic; concept head



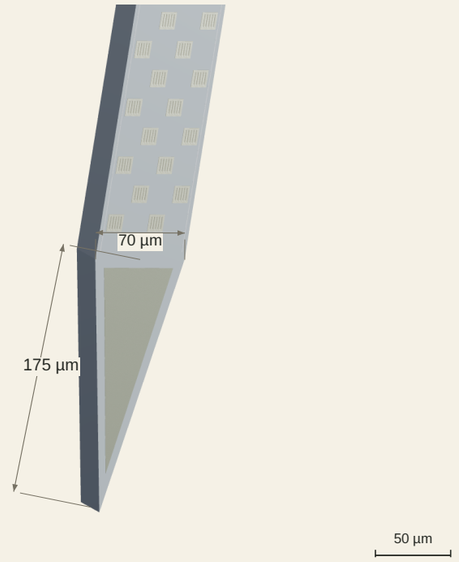
C Side elevation

Orthographic



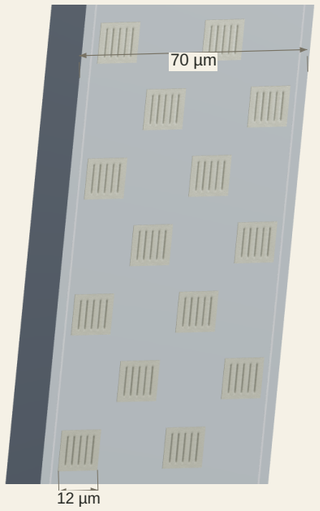
D Distal tip

Chisel + reference electrode



E Electrode-array detail

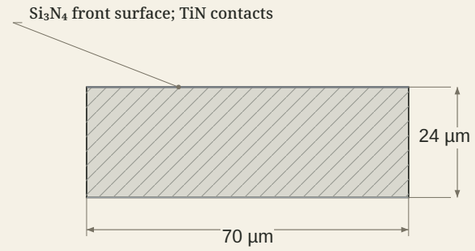
Four staggered column positions / two sites per row



F Transverse section

Outer envelope from the exported CAD solid

The cut is through a populated row.
No unverified internal stack is drawn.
Surface coatings are labeled, not thickened.



Silicon / silicon-oxide sidewall; SiO_2 back surface.

Contact and coating thicknesses are not verified.

20 µm

MODEL NOTES

01 First site centre = 200 µm from tip: assumed absolute origin; not a hardware channel map.
02 Straight 175 µm taper gives 22.62° included angle. The datasheet angle is approximate.
03 Slot dimensions, relief and pore morphology are illustrative; no measured porosity is claimed.
04 All shaded views use the delivered meshes. Each detail has its own scale bar.
05 No internal CMOS routing, tolerances, electrical function or fabrication readiness is implied.

EVIDENCE

IMEC / Neuropixels 1.0 manufacturer datasheet
13 February 2020, pp. 3–4: dimensions, materials, SEM.
Official Neuropixels support and product imagery.
Full source URLs and assumptions: specification.json.
An independent reconstruction; not manufacturer CAD.

NEUROPIXELS 1.0 / ENGINEERING REFERENCE

CATEGORY / MEMS • NEURAL INTERFACE

UNITS / mm & µm PROJECTION / ORTHOGRAPHIC

REFERENCE — NOT FOR FABRICATION